TELEMEDICINE APP

A Digital Healthcare Solution
for Remote Access

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Telemedicine app

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PROJECT GUIDE

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Abstract

The telemedicineapp enables remotemedicalconsultations using digital tools. It serves patients by offering health services without physical visits, improvingaccessin underserved areas. Key features includeappointment scheduling, and electronic health recordintegration.The app is built with user-friendly interfacesandrobustsecurity measures to protect patient data.

Introduction

- This presentation covers the essential aspects of a telemedicine app designed to provide remote healthcare access. It highlights how technology bridges the gap between patients and healthcare providers, overcoming geographic and time barriers. The app promotes efficiency, accessibility, and convenience in medical consultations and monitoring.

Existing System

- Patients travel long distances to reach hospitals.
- Specialistdoctors are not available in villages.
- Poor internet connectivity stops online consultation.
- Medical records are kept on paper and may get lost.
- Treatment is delayed due to travel and waiting time

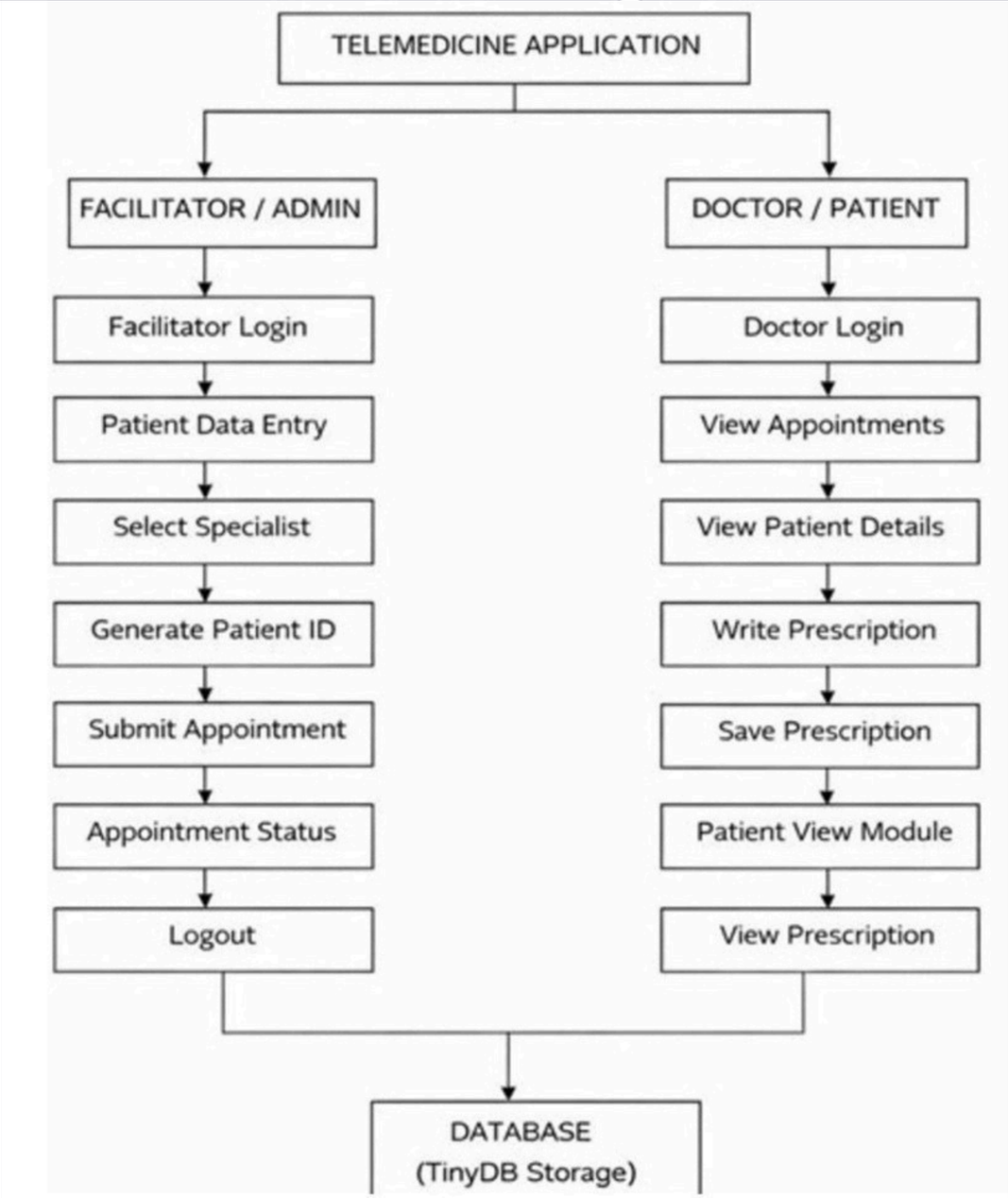
Proposed System

- Facilitator collects patient details in the village.
- Patients consult doctors from a stable network area.
- Doctors provide advice through the telemedicine app.
- Prescriptions are given digitally to patients.
- Saves time, money, and improves rural healthcare

Design

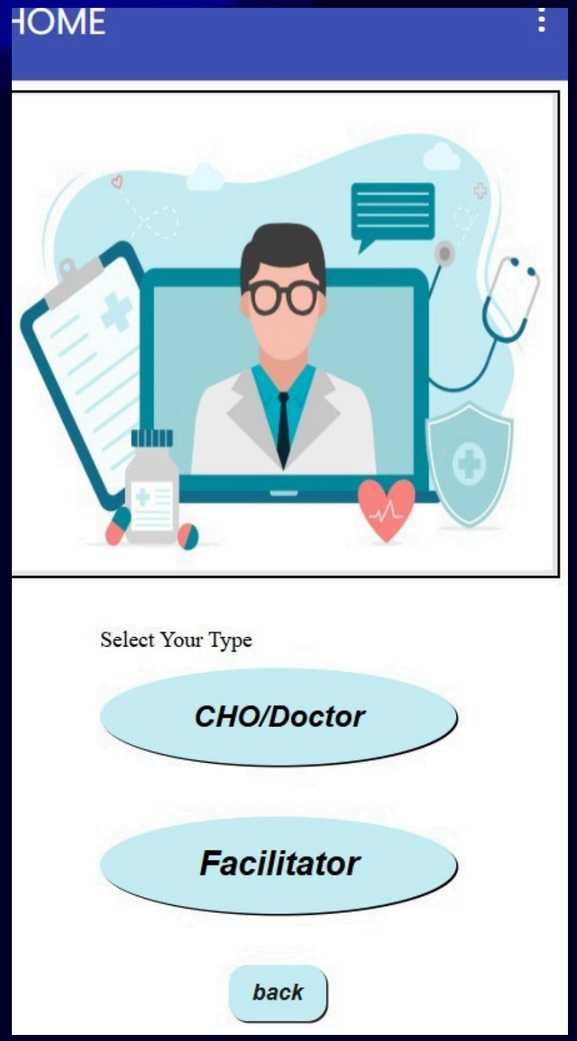
1. The Telemedicine App follows a client–server architecture to ensure medical assistance.
2. The frontend is developed using MIT App Inventor to provide a simple, user- friendly mobile interface.
3. TinyDB Realtime Database is used as the backend to securely store . patient details, doctor profiles, appointments, and prescriptions.
4. The system allows patients to register, login, book appointmentrs
5. Appointment management ensures real-time availability, secure data handling, and confirmationy

Block diagram

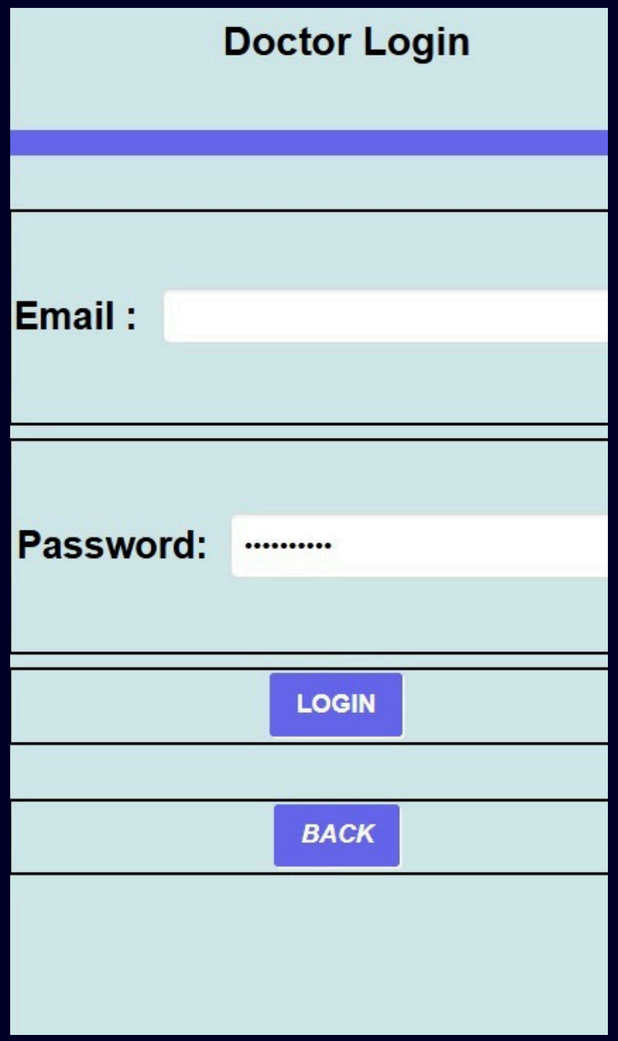


Implementation

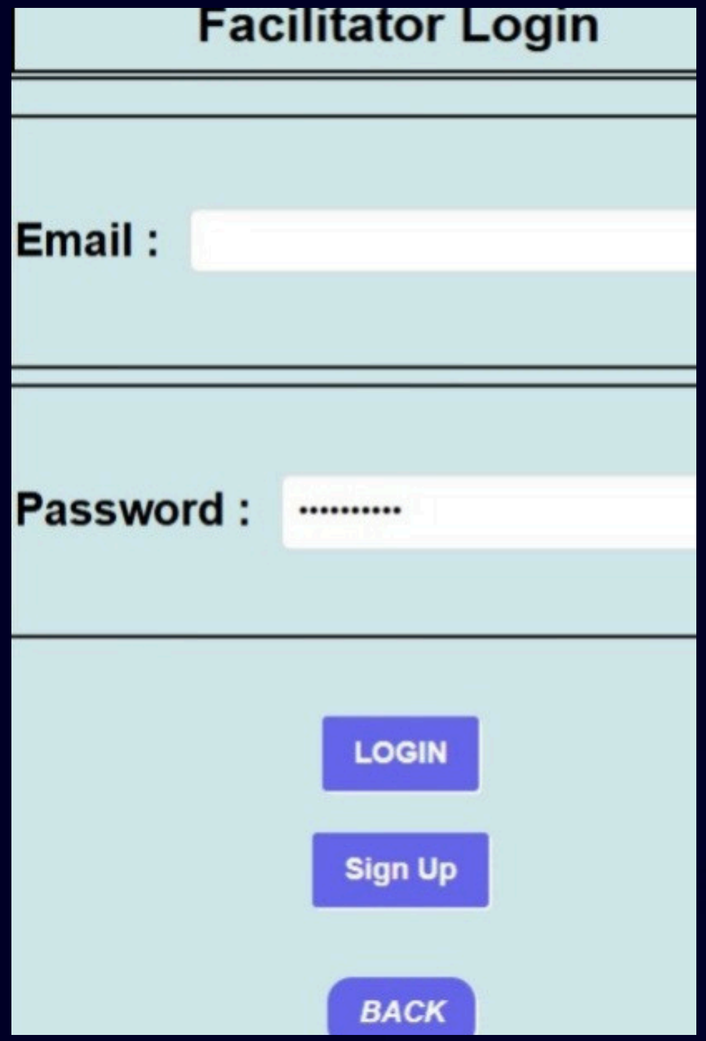
- User Authentication:Patients and doctors register and login using email and password authentication .
- Patient Profile Management: Users enter and update personal details such as name, age, gender, contact number, and medical history.
- Doctor & Specialist Selection: Patients select required medical specialization and view available doctors.
- Appointment Booking Module: Users choose availale doctors and appointments are stored in Tiny DB.



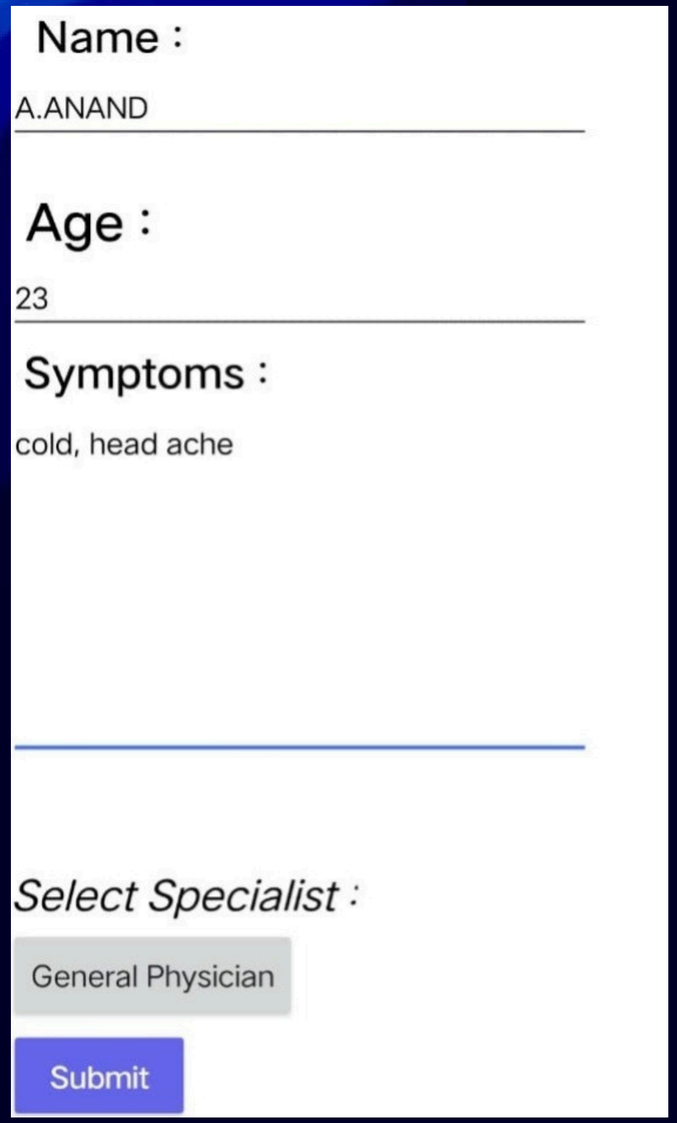
Role selection



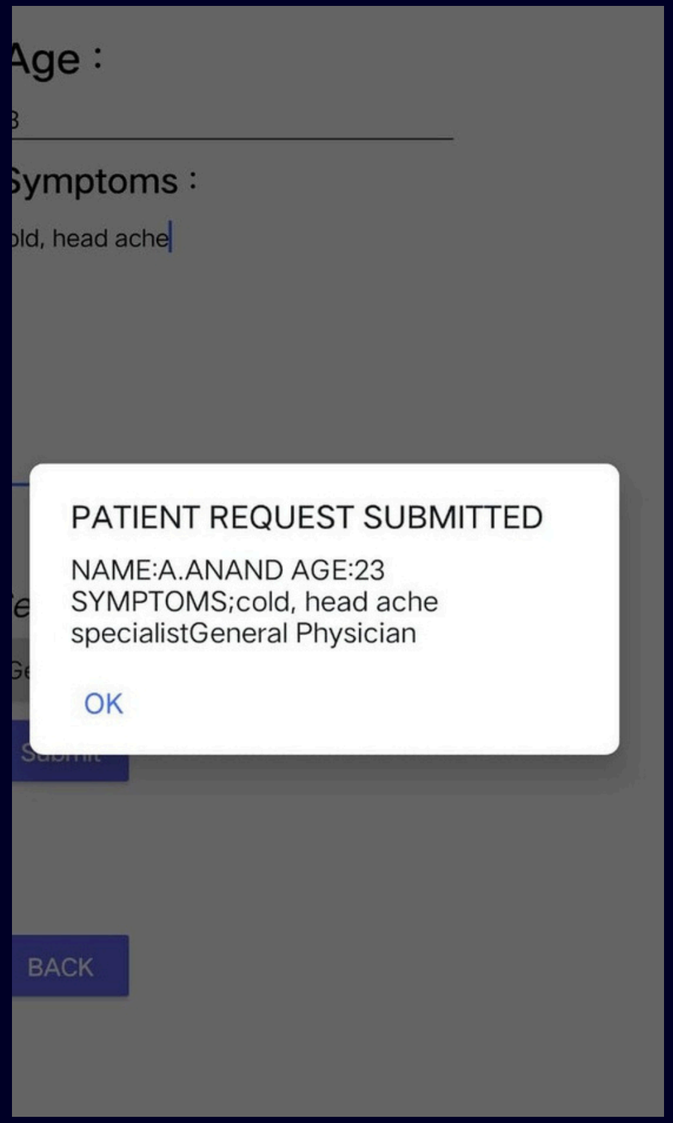
Doctor login



Facilitator login

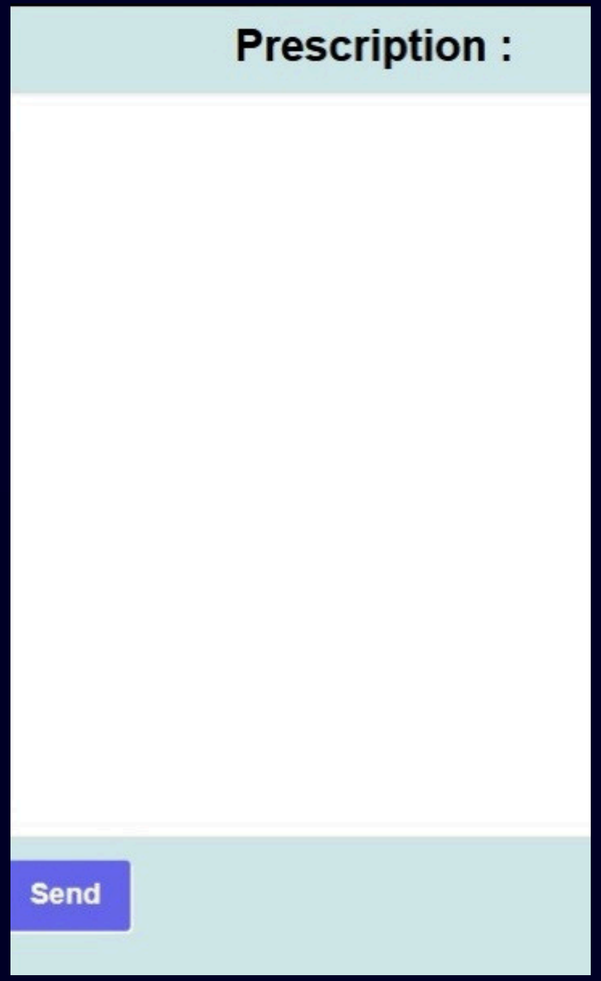


Patient request

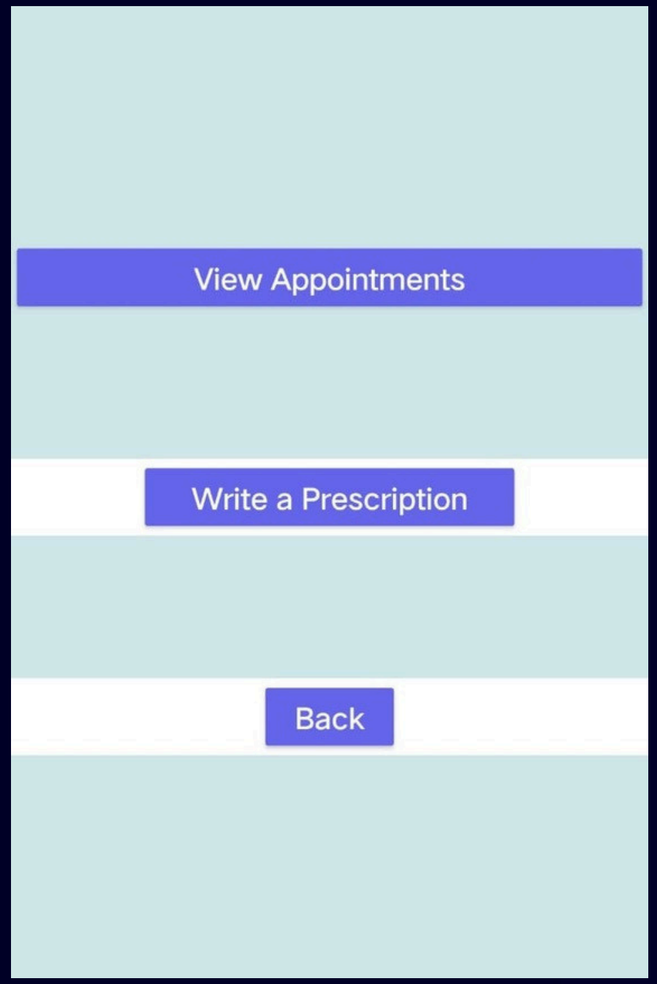


Confirmation

View prescription



Prescription



Doctor dashboard

Testing

- Performed login testingforfacilitatoranddoctor authentication.
- Verified patient data storage and retrieval using TinyDB.
- Tested prescription generation and Patient ID-based access.
- Confirmed system stability through unit, functional, and system testing.

Result

- The system enables fast and paperless consultation between facilitator, doctor, and patient.
- Accurate data storage and retrieval using TinyDB ensures no loss of patient or prescription records.
- Doctors can generate digital prescriptions, and patients can access them using Patient ID, reducing delays.
- The app is low-cost, user-friendly, and suitable for rural areas, developed using MIT App Inventor from Massachusetts Institute of Technology.

CONCLUSIONS

This telemedicine app offers a scalable and secure platform to connect patients and healthcare providers remotely. Through careful development and ongoing enhancements, it meets current healthcare demands and anticipates future needs. It stands as a vital tool in modern healthcare, promoting accessibility, convenience, and improved medical outcomes worldwide.

Future Scope and Enhancements

Future enhancements aim to incorporateAI-powered diagnostics, advanced symptom trackers,and integration with wearable devices for continuous health monitoring. Expanding multilingual support and improving interoperability with healthcare systems will broaden the app’s reach and effectiveness, ensuring enhanced patient outcomes and provider efficiency.

BIBLIOGRAPHY

This project is developedby referringtoreliablesources on telemedicine systems and mobile health technologies. The following references provided guidance for designing and building the application using MIT App Inventor.

- World Health Organization – Reports on telemedicine and digital health services
- Ministry of Health and Family Welfare – Telemedicine Practice Guidelines.
- J Garofolo et al., Mobile Health (mHealth) Technologies: Applications and Services – Reference for healthcare app
- Daniel Noel & Paul C. Deitel, Android App Development for Beginners – Basics useful for app design.

Thank You